

Does Safety Survive Encoding?

Certiability Limits of Learned Representations

Research Proposal — v2

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Abstract

Safety certificates are increasingly defined in learned latent spaces. Every such result I am aware of assumes that the representation transports the safety specification faithfully, via an ε -conjugacy or bounded-transport condition whose constant is asserted rather than established. This proposal asks the prior question — *when can a safety property be represented faithfully in a latent space at all?* — and answers it with a representation-level quantity, the *safety defect* ε_{sep} . Three results follow. (i) ε_{sep} is *exactly* the margin any sound latent certificate must sacrifice; it is a lower bound on the conservatism of every downstream method and is independent of the certificate class. (ii) Expected representation losses cannot control it: arbitrarily good reconstruction and prediction error coexist with an $O(1)$ defect, so the resulting certification gap is invisible to every metric the field reports. (iii) The property that does control it is not invertibility but *safety-faithfulness* — the specification, not the state, must survive compression — which is trainable and admits a dimension-free probabilistic certificate. The decisive question is empirical, and a completed pilot over 48 ordinarily trained encoders (§7) answers it: at *matched* validation reconstruction ($R^2 \approx 0.977$ in both arms) the defect differs by $7.8\times$, and in the best-reconstructing configuration ($R^2 = 0.992$) a state with three safety radii of margin is latent-indistinguishable from a violating one. Control barrier functions appear as the downstream demonstration, not as the subject.

1 Thesis and contributions

Latent world models and latent diffusion policies now routinely carry safety filters defined on the latent state [1, 2, 4]. The guarantees are always conditional: *given* an encoder satisfying an ε -forward-conjugacy condition, latent safety implies state safety. The most rigorous instance, set-based verification by zonotope propagation [3], notes that the required constants “may be difficult to obtain precisely” and that its certificates are conservative. Empirically, latent certificates are observed to pass while task-level safety fails [5], with no account of why.

Thesis. *Whether a safety property survives encoding is a measurable property of the representation, it exactly determines the best certificate any method can produce, it is not delivered by any objective currently used to train representations, and it is restored by a condition on the specification rather than on the state.*

The reframing is the contribution: the field asks “given a good representation, can we certify safety?”; this asks “when is safety representable?”. That question is broader than CBFs, which is what makes it an ICML paper rather than another certificate construction. Concretely:

- C1.** The safety defect ε_{sep} (Def. 4) and an exact soundness–completeness theorem (Thm. 5): ε_{sep} is the minimal certifiable margin, attained by an explicit certificate.
- C2.** A geometric characterisation of when a defect arises (Prop. 7): the encoder may compress along level sets of h and nowhere else.
- C3.** A separation (Thm. 8): expected reconstruction and prediction error do not control ε_{sep} , so the certification gap of C1 is invisible to reported metrics.

- C4.** Safety-faithfulness (Def. 10), a trainable condition that controls ε_{sep} , with an estimation taxonomy (Props. 13–15) including a *dimension-free* probabilistic certificate.
- C5.** An empirical study establishing that the phenomenon occurs in ordinarily trained encoders (§7), and that the intervention removes it without destroying utility.

2 Setup

Assumption 1 (Plant and specification). $\mathcal{X} \subseteq \mathbb{R}^n$ with $\dot{x} = f(x) + g(x)u$, $u \in \mathcal{U}$, f, g locally Lipschitz. The safe set is $\mathcal{C} = \{x : h(x) \geq 0\}$ for L_h -Lipschitz, C^1 h with $\nabla h \neq 0$ on $\partial\mathcal{C}$. Fix a compact region of interest $\mathcal{K} \subseteq \mathcal{X}$ and a distribution ρ on $\mathcal{K}$ (the data or reachable-state distribution).

Assumption 2 (Representation). C^1 encoder $E : \mathcal{X} \rightarrow \mathcal{Z} \subseteq \mathbb{R}^d$ and decoder $D : \mathcal{Z} \rightarrow \mathcal{X}$, $d \leq n$.

Definition 3 (Soundness, completeness). A latent certificate is $h_z : \mathcal{Z} \rightarrow \mathbb{R}$ with $\mathcal{C}_z = \{h_z \geq 0\}$. It is *sound* on $\mathcal{K}$ if $E^{-1}(\mathcal{C}_z) \cap \mathcal{K} \subseteq \mathcal{C}$, and *m-complete* if $\{x \in \mathcal{K} : h(x) \geq m\} \subseteq E^{-1}(\mathcal{C}_z)$.

Soundness alone is vacuous ($h_z \equiv -1$). The content is the tradeoff, and one quantity governs it.

Definition 4 (Safety defect).

$$\varepsilon_{\text{sep}}(E) := \sup \{ h(x) : x \in \mathcal{K}, \exists x' \in \mathcal{K}, E(x') = E(x), h(x') < 0 \} \vee 0, \quad (1)$$

the largest margin attained by a safe state sharing a code with an unsafe one. Its δ -robust version $\varepsilon_{\text{sep}}^\delta$ replaces $E(x') = E(x)$ with $\|E(x') - E(x)\| \leq \delta$.

3 C1 — The defect is exactly the minimal certifiable margin

Theorem 5 (Exact soundness–completeness tradeoff). *Under Assumptions 1–2 with $\mathcal{K}$ compact: (i) no sound latent certificate is m-complete for any $m \leq \varepsilon_{\text{sep}}(E)$; and (ii) $h_z^*(z) := \inf\{h(x) : x \in \mathcal{K}, E(x) = z\}$ is sound and m-complete for every $m > \varepsilon_{\text{sep}}(E)$. Hence the minimal certifiable margin is $m^*(E) = \varepsilon_{\text{sep}}(E)$.*

Proof. (i) Suppose h_z is sound and m-complete, $m \leq \varepsilon_{\text{sep}}$. Pick $x, x' \in \mathcal{K}$ with $E(x) = E(x')$, $h(x) \geq m$, $h(x') < 0$. Completeness gives $E(x) \in \mathcal{C}_z$; since $E(x') = E(x)$, soundness gives $h(x') \geq 0$, a contradiction. (ii) If $h_z^*(E(x)) \geq 0$ then every state in the fiber satisfies $h \geq 0$, so h_z^* is sound. If $h(x) > \varepsilon_{\text{sep}}$ then no unsafe state lies in the fiber of $E(x)$, so $h_z^*(E(x)) \geq 0$; compactness and continuity make the infimum attained. $\square$

Corollary 6 (Irreducible conservatism). *Any framework transporting safety from latent to state space under an ε -conjugacy or bounded-transport assumption must have $\varepsilon \geq \varepsilon_{\text{sep}}(E)$. The conservatism observed in [3] is bounded below by a property of the learned encoder, not by the sharpness of the set propagation.*

Proposition 7 (Safety-relevant collapse). *Let $x_0 \in \partial\mathcal{C} \cap \text{int } \mathcal{K}$. If there is $v \in \ker dE_{x_0}$ with $\nabla h(x_0)^\top v = \mu \neq 0$, then $\varepsilon_{\text{sep}} \geq c|\mu|r + o(r)$ for $B(x_0, r) \subseteq \mathcal{K}$. Conversely $\varepsilon_{\text{sep}} = 0$ requires $\nabla h(x) \perp \ker dE_x$ near $\partial\mathcal{C} \cap \mathcal{K}$: the specification must factor through the encoder to first order.*

Theorem 5 is deliberately certificate-class-independent: it constrains neural barriers, polynomial certificates, zonotope propagation and CBF-QPs alike, which is what prevents the paper from reading as another certificate construction.

4 C3 — Representation losses do not control the defect

Theorem 8 (No transfer from expected loss). *Let $n \geq 2$ and let ρ have bounded density on $\mathcal{K}$. For every $\eta > 0$ there is a C^∞ pair (E_η, D_η) with $\mathbb{E}_\rho \|D_\eta(E_\eta(x)) - x\|^2 \leq \eta$ and one-step latent prediction error $\leq \eta$, yet $\varepsilon_{\text{sep}}(E_\eta) \geq c_0 > 0$ with c_0 depending only on h and $\mathcal{K}$. Hence no ψ with $\psi(0^+) = 0$ satisfies $\varepsilon_{\text{sep}} \leq \psi(\mathcal{L}_{\text{train}})$.*

Proof sketch. Let $T_{\sigma,\ell} \subseteq \mathcal{K}$ be a tube of transverse radius σ and length ℓ aligned with ∇h and straddling $\partial\mathcal{C}$. Let E_η be the identity outside $T_{\sigma,\ell}$ and a smooth reflection about the tube’s midpoint inside, identifying x with its mirror image; mollify to obtain smoothness. Both losses are supported on $T_{\sigma,\ell}$, of ρ -mass $O(\sigma^{n-1}\ell)$, hence vanish as $\sigma \rightarrow 0$ at fixed ℓ ; the identified pairs straddle $\partial\mathcal{C}$ along ∇h over length ℓ , so $\varepsilon_{\text{sep}} \geq \frac{1}{2}\ell \inf_T \|\nabla h\| =: c_0$, independent of σ . $\square$

Remark 9 (What is and is not being claimed). That an expectation fails to control a supremum is not itself interesting, and the paper must not be sold that way. The claim is sharper on three counts. First, the mismatch does not merely loosen a bound — by Theorem 5 it produces an *irreducible* certification gap of size exactly ε_{sep} , which no downstream method can recover. Second, the gap is invisible to every quantity currently reported for these models (reconstruction, ELBO, rollout error, downstream return), so practitioners have no instrument that would reveal it. Third, it is cheap to remove (§5), which makes the observation actionable rather than merely cautionary. Theorem 8 is therefore the *existence* half of the story; §7 carries the burden of showing the phenomenon is typical rather than pathological, and the project’s significance rests there.

5 C4 — Safety-faithfulness, and how to certify it

Invertibility restores $\varepsilon_{\text{sep}} = 0$ but is far stronger than necessary: an encoder may discard everything the specification does not see.

Definition 10 (κ -safety-faithful). $|h(x) - h(x')| \leq \kappa \|E(x) - E(x')\|$ for all $x, x' \in \mathcal{K}$; equivalently h factors through E with modulus κ .

Theorem 11 (Certifiability under faithfulness). *If E is κ -safety-faithful then $\varepsilon_{\text{sep}} = 0$, $\varepsilon_{\text{sep}}^\delta \leq \kappa\delta$, and h_z^* is κ -Lipschitz, sound and m -complete for all $m > 0$. A latent margin m_z certifies state margin m_z/κ .*

Proposition 12 (Compression remains possible). *Let $h = \varphi \circ \pi$ for a submersion $\pi : \mathcal{X} \rightarrow \mathcal{Q}$ onto a q -dimensional quotient and Lipschitz φ . If $E = \psi \circ \pi$ with ψ injective and co-Lipschitz on $\pi(\mathcal{K})$, then E is κ -safety-faithful with $\kappa = \text{Lip}(\varphi)\text{Lip}(\psi^{-1})$ — although E is non-injective on $\mathcal{X}$ whenever $q < n$. The minimal certifiable latent dimension is therefore q , the dimension of the safety-relevant quotient, not n .*

Safety-faithfulness is trainable as a pairwise penalty,

$$\mathcal{R}_\kappa(E) = \mathbb{E}_{x,x' \sim \rho} [(|h(x) - h(x')| - \kappa \|E(x) - E(x')\|)_+^2]. \quad (2)$$

Certifying it is a separate matter, and the honest answer is a taxonomy rather than a single bound.

Proposition 13 (Sound but dimension-limited). *If (2) holds on all pairs of an r -cover of $\mathcal{K}$ and E is L_E -Lipschitz, then $\varepsilon_{\text{sep}} \leq 2L_h r + \kappa L_E r$. The cover needs $N = O((\text{diam } \mathcal{K}/r)^m)$ points with m the intrinsic dimension of $\text{supp } \rho$. This is a genuine soundness guarantee and is appropriate for low-intrinsic-dimension systems and for boundary-localised verification; it is not a scalable instrument, and the proposal does not claim otherwise.*

Definition 14 (Probabilistic defect). For $\alpha \in (0, 1)$ let $V(x) := h(x)\mathbf{1}\{x \text{ is } \delta\text{-confounded}\}$, where x is δ -confounded if some x' with $h(x') < 0$ has $\|E(x') - E(x)\| \leq \delta$. Set $\varepsilon_{\text{sep}}^{\delta, \alpha} :=$ the $(1 - \alpha)$ -quantile of V under ρ .

Proposition 15 (Dimension-free probabilistic certificate). *Let $V_1, \dots, V_N$ be i.i.d. under ρ with order statistics $V_{(1)} \geq \dots \geq V_{(N)}$. For any k , with probability at least $1 - \sum_{i < k} \binom{N}{i} \alpha^i (1 - \alpha)^{N-i}$ we have $\rho(V > V_{(k)}) \leq \alpha$. In particular $N \geq \log(1/\delta_{\text{conf}})/\alpha$ samples certify $\varepsilon_{\text{sep}}^{\delta, \alpha} \leq V_{(1)}$ with confidence $1 - \delta_{\text{conf}}$. The bound is distribution-free and independent of both n and d .*

Remark 16 (Reading the taxonomy). Proposition 15 bounds the *probabilistic* defect and is the instrument that scales; Proposition 13 bounds the *exact* defect and does not. A third mode — nearest-neighbour search over a held-out sample — gives an empirical *lower* bound, which is what §7 uses, since falsifying an existing practice needs only a lower bound. Stating which of the three a given claim rests on is, I think, the correct way to handle the scalability objection: the exact defect is a verification problem and inherits verification’s costs, while the quantity a deployed system actually needs is the probabilistic one.

6 Architectural consequences

Theorem 11 and Proposition 12 give two routes, and the distinction between them is the useful statement — not a claim that invertibility is necessary.

Specification-aware. If h is known at training time, (2) yields faithfulness at the minimal latent dimension q of Proposition 12. This is the weakest sufficient condition and the preferred route.

Specification-agnostic. If h is unknown, unavailable, or expected to change, faithfulness cannot be trained for. An encoder that is a C^1 diffeomorphism onto its image is κ -safety-faithful *simultaneously for every* L_h -Lipschitz specification, with $\kappa = L_h / \inf_x \sigma_{\min}(\mathrm{d}E_x)$. Flow-based representations deliver this by construction, and $\sigma_{\min}$ is accessible in practice (tracked log-determinants; spectral normalisation of the velocity field; a Grönwall bound on the Jacobian singular values for continuous-time flows).

So invertibility buys *specification-independence* at the price of κ and of forgoing compression below n ; safety-faithfulness buys certifiability for one known h at dimension q . This is an architectural consequence of the theory rather than its headline.

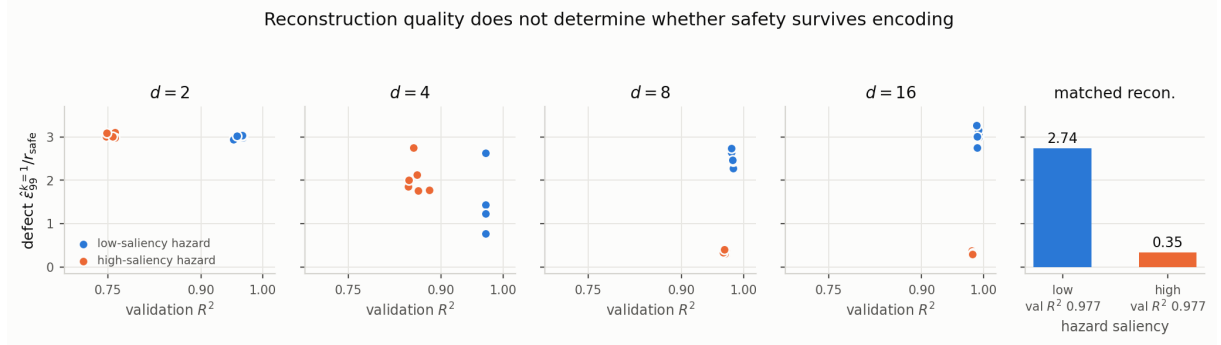
7 Empirical programme

The theory is only as interesting as the answer to one question: *do ordinarily trained encoders have defects?* If they do not, Theorem 8 is an existence counterexample and the significance collapses. This is therefore the first experiment, not the last.

Pilot: setup. A deliberately mundane task with no folds induced. State $s = (\text{agent}, \text{obstacle}) \in \mathbb{R}^4$; observation a 16×16 image in which the agent is a large bright blob and the obstacle a small one; specification $h(s) = \|\text{agent} - \text{obstacle}\| - r_{\text{safe}}$, $r_{\text{safe}} = 0.18$. The hazard is thus reconstruction-cheap but safety-decisive. This is the realistic mechanism and it needs no adversary: *safety-relevant factors need not be reconstruction-relevant factors*, and a hazard that occupies few pixels is the ordinary case in pixel-based control. Standard β -VAEs with a latent transition model were trained over $d \in \{2, 4, 8, 16\}$, $\beta \in \{0, 10^{-3}\}$, two obstacle saliencies and three seeds (48 models; a 96-model grid with $\beta \in \{0, 10^{-4}, 10^{-3}, 10^{-2}\}$ gives the same picture, and β was not an important factor).

Two controls make any loss attributable to the representation rather than to the sensor: a *supervised observability ceiling* — a network reading h directly from the image, which achieves $R^2(h \mid \text{image}) = 0.993\text{--}0.998$ in every condition, so the hazard is plainly visible — and a *capacity-matched nonlinear probe* reading h from the latent code. The defect estimator is rank-based and therefore free of any collision radius: a safe held-out state is *confounded* if its nearest held-out latent neighbour is unsafe, and $\hat{\varepsilon}_{99}^{k=1}$ is the 99th percentile of h over confounded states. (A δ -ball estimator was tried first and rejected: it is knife-edged in δ . The two agree at $r = +0.83$.) The estimator is a *lower* bound on ε_{sep} , which is all a falsification needs.

Pilot: result. Defects are large, and reconstruction quality does not predict them. The two saliency conditions reach *indistinguishable* validation reconstruction — $R^2 = 0.9770 \pm 0.0108$ versus 0.9766 ± 0.0062 — and differ in defect by $7.8\times$: median $\hat{\varepsilon}_{99}^{k=1}/r_{\text{safe}}$ of 2.74 versus 0.35. Within the low-saliency condition, the correlation between validation R^2 and the defect is $+0.17$, and between one-step prediction R^2 and the defect $+0.07$: neither reported metric carries information about it. The strongest single configuration is the sharpest case — $d = 16$, $\beta = 0$: validation $R^2 = 0.992$, prediction $R^2 = 0.907$, latent probe $R^2(h \mid z) = 0.96$, and yet a defect of $3.08 r_{\text{safe}}$, with 6.6% of safe held-out states having an unsafe nearest latent neighbour. A state with three safety radii of margin is latent-indistinguishable from a violating one in a model that reconstructs at 99.2% and whose latent visibly encodes h .



Remark 17 (Two readings). First, the last observation is Theorem 8 recurring one level down: probe R^2 is itself an averaged criterion, so a latent that encodes h well *on average* still collides at the boundary. The L^2/L^∞ gap is not an artifact of the pixel objective; it reappears wherever a mean-squared quantity is used to argue about a worst-case property. Second, the k -NN latent certificate almost never *false-certifies* (median rate 0): it stays sound and pays in completeness — exactly the trade Theorem 5 says is forced. The failure mode of latent safety is not silent unsoundness but silent, and large, conservatism.

Pilot: what it does not yet show. The system is synthetic and low-dimensional; the estimator lower-bounds the defect; and the mechanism exercised here is hazard saliency, with posterior collapse via β playing little role. E1 is therefore not optional — the claim “this happens in ordinary practice” requires standard pixel world models, not a constructed task, however mundane.

Planned. (E1) the same measurement on standard pixel world models (pendulum, cart-pole, car racing); (E2) the money figure — validation loss against false-certification rate across many trained encoders, with (2)-regularised encoders overlaid on a different frontier; (E3) specification-aware versus specification-agnostic routes at matched reconstruction quality; (E4) certified margin versus d and κ , locating the safety-relevant dimension q empirically; (E5) a latent CBF-QP on a driving or manipulation task, as the downstream demonstration.

8 Risks

Defects are rare in practice. The decisive risk, addressed first (§7). Fallback: C1, C4 and the estimation taxonomy stand independently as the first quantitative account of the compression–certifiability frontier.

“An expectation does not control a supremum.” Anticipated reviewer objection; Remark 9 states the actual claim, and the empirical section is what answers it.

Exact defect estimation is expensive. Acknowledged and structured as the taxonomy of Props. 13–15 rather than concealed.

h is learned. Then ε_{sep} is relative to the learned specification and composes with its estimation error — a limitation to state, not to fix.

9 Scope and timeline

Deliberately excluded from the first paper: the composition theorem aggregating ε_{sep} , conjugacy error and the class- $\mathcal{K}$ gain into a single CBF tightening, and the viability-theoretic question of how much of the viability kernel survives encoding (a two-sided bound on $d_H(\text{Viab}(\mathcal{C}), D(\text{Viab}_z(\mathcal{C}_z)))$ over a horizon T). Both are attractive and both dilute a crisp thesis; they belong to papers 2 and 3.

Months	Focus	Deliverable
0–1	pilot	§7 (<i>done</i>): defect is large and reconstruction-invisible
1–4	C1–C3	Thms. 5, 8, Prop. 7; defect estimator; E1
4–7	C4–C5	(2), Props. 12–15; E2–E4
7–9	—	Paper 1 (ICML/NeurIPS)
9–15	downstream	composition theorem, E5; Paper 2 (CDC/L4DC)
15–24	outlook	viability under encoding; Paper 3

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